SORTING PAIRS IN BINS: EXACT VALUES, A CONJECTURED CLOSED FORM, AND PARTIAL PROGRESS ON A PUZZLE OF WINKLER

ABSTRACT. We study the *Sorting Pairs in Bins* puzzle, listed among the unsolved puzzles in Peter Winkler's *Mathematical Puzzles: A Connoisseur's Collection*: n bins are arranged in a row, the i-th bin initially containing two balls labeled $n+1-i$; a move consists of swapping one ball from bin i with one ball from bin $i+1$; the goal is to determine $T(n)$, the minimum number of moves needed to bring bin i to contain two balls labeled i for every i. We report exact values of $T(n)$ for $n \le 7$, obtained by exhaustive and bidirectional breadth-first search, and conjecture the closed-form formula

$$T(n) = \left\lfloor \frac{n^2}{2} \right\rfloor + S(n), \qquad S(n) = \sum_{k=2}^{n} \left\lfloor \frac{k}{3} \right\rfloor,$$

which we verify exactly for $1 \le n \le 7$ and for which we give an elementary closed form. We prove the baseline lower bound $T(n) \ge \lfloor n^2/2 \rfloor$ via a displacement potential, establish an exact accounting identity decomposing $T(n)$ into a displacement term and an "excess" contributed by non-progressing moves, and reformulate the problem as a weighted reduced-word problem for the longest element of S_{2n}, in which we prove a new structural lemma governing the temporal order of three-wire crossings. We rigorously establish that several natural proof strategies — single-wire arguments, positional-parity arguments, local aggregate potentials analyzed via linear programming over the full reachable state graph, pure local-greedy constructions, and a natural bin-removal induction — are all individually insufficient to prove the conjectured excess term, and we document each insufficiency with an explicit witnessing example. We introduce an auxiliary "wall-crossing" subproblem for which we prove an exact linear formula, and report a striking, data-grounded "50% absorption" phenomenon in a nested generalization of it that is in direct numerical tension with the main conjecture at $n = 8$. We close with a precise statement of this open discrepancy and a list of open problems.

1. INTRODUCTION

In *Mathematical Puzzles: A Connoisseur's Collection* [1], Peter Winkler lists, among a handful of puzzles he was unable to resolve at the time of writing, the following.

> **Sorting Pairs in Bins.** Standing in a row are n bins, the i-th bin containing two balls numbered $n+1-i$. At any time, you may swap two balls from adjacent bins. How many swaps are needed to get every ball into the bin carrying its number?

We interpret "swap two balls from adjacent bins," consistently with the accompanying figure in [1], as: choose one ball from bin i and one ball from bin $i+1$, for some i, and exchange them. Denote by $T(n)$ the minimum number of such swaps required to reach the target configuration, in which bin i contains two balls labeled i for every i.

This puzzle sits in the same family as the *Malicious Maître D'* problem, also from [1], whose adaptive variant was recently resolved by Madhavarapu [2], building on work of Acton, Kotek, Miller, and others. That paper's methodology — exact combinatorial analysis of a discrete optimization puzzle from Winkler's book, informed by extensive computational experimentation — served as a model for the present investigation.

Unlike the Malicious Maître D' problem, we have not been able to obtain a complete solution to Sorting Pairs in Bins. What we report here is, we believe, still of value: an exact conjectural formula supported by full verification for $n \le 7$; several genuine theorems constraining any eventual proof; a systematic account of natural proof strategies that we have rigorously ruled out, together

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with explicit witnessing counterexamples, so that future investigators need not repeat this work; and a striking auxiliary result whose naive extrapolation directly conflicts with our main conjecture at the very next unverified value, $n = 8$ — a discrepancy we present honestly as unresolved, since at the time of writing $T(8)$ has not been independently computed.

We hope this paper is useful in two ways: as a source of exact data and rigorous partial results for anyone attempting to complete the proof, and as a case study in how far one can push a computational-plus-structural investigation of a discrete optimization puzzle before hitting a genuine wall, together with a candid account of exactly where that wall is.

1.1. **Organization.** Section 2 sets notation and reports the exact values of $T(n)$ for $n \leq 7$. Section 3 states the main conjecture and gives an elementary closed form for its correction term $S(n)$. Section 4 proves the baseline displacement lower bound and an exact accounting identity. Section 5 develops a reformulation of the problem in terms of reduced words for the longest element of a symmetric group, proves a new lemma about three-wire crossings, and explains precisely why this reformulation, as far as we have been able to push it, does not yield the full conjectured bound. Section 6 catalogues further proof strategies we have ruled out, each with an explicit counterexample. Section 7 introduces an auxiliary subproblem for which we prove an exact formula, and reports the 50%-absorption phenomenon and its unresolved tension with the main conjecture at $n = 8$. Section 8 collects open problems.

2. Preliminaries and exact values

Definition 2.1. For a positive integer n, an *n-bin configuration* is an assignment, to each of n bins indexed $1, \ldots, n$, of an unordered pair (with repetition) of labels from $\{1, \ldots, n\}$, such that each label $1, \ldots, n$ occurs in exactly two bins total (counting multiplicity). The *start configuration* has bin i containing $\{n+1-i, n+1-i\}$ for each i. The *target configuration* has bin i containing $\{i, i\}$ for each i.

A *move* at boundary i (for $1 \leq i \leq n-1$) selects one ball from bin i and one ball from bin $i+1$ and exchanges their bins. Let $T(n)$ denote the minimum number of moves required to transform the start configuration into the target configuration.

We computed $T(n)$ exactly for $n \leq 7$ using exhaustive breadth-first search over the full reachable configuration graph, cross-validated by an independent bidirectional BFS implementation. Table 1 lists the results together with the size of the reachable state space explored.

n	$T(n)$	reachable states
1	0	1
2	2	3
3	5	21
4	10	282
5	15	6,210
6	23	202,410
7	31	$\sim 8 \times 10^6$ (bidirectional meeting point)

Table 1. Exact values of $T(n)$, $1 \leq n \leq 7$.

Attempts to extend Table 1 to $n = 8$ by the same exhaustive methods, even after substantial optimization (compact state encoding, open-addressing hash tables, bidirectional search), have not completed within the memory and time available to us; reachable-state counts in excess of 10^8 were reached without termination. We regard $T(8)$ as currently unknown by exhaustive computation, and we return to why this matters in Section 7.

3. The conjectured formula

Conjecture 3.1. *For all $n \geq 1$,*

$$T(n) = \left\lfloor \frac{n^2}{2} \right\rfloor + S(n), \qquad S(n) := \sum_{k=2}^{n} \left\lfloor \frac{k}{3} \right\rfloor .$$

Conjecture 3.1 is verified exactly against every entry of Table 1. It is not, at present, proved; we regard establishing it as the central open problem of this line of work.

The correction term $S(n)$ admits a fully elementary closed form.

Theorem 3.2. *Write $n = 3q + r$ with $q = \lfloor n/3 \rfloor$ and $r \in \{0, 1, 2\}$. Then*

$$S(n) = \frac{3q(q-1)}{2} + q(r+1) = \begin{cases} \dfrac{q(3q-1)}{2}, & r = 0, \\ \dfrac{q(3q+1)}{2}, & r = 1, \\ \dfrac{3q(q+1)}{2}, & r = 2. \end{cases}$$

Proof. Since $\lfloor 0/3 \rfloor = \lfloor 1/3 \rfloor = 0$, we have $S(n) = \sum_{k=0}^{n} \lfloor k/3 \rfloor$. For $k = 0, \ldots, n$, the value $\lfloor k/3 \rfloor$ equals j for exactly the three consecutive integers $k = 3j, 3j+1, 3j+2$ whenever $3j + 2 \leq n$, and additionally for $k = 3q, \ldots, 3q + r$ ($r + 1$ values) at the top when $r < 2$. Summing,

$$\sum_{k=0}^{n} \left\lfloor \frac{k}{3} \right\rfloor = 3 \sum_{j=0}^{q-1} j \; + \; q(r+1) = 3\binom{q}{2} + q(r+1) = \frac{3q(q-1)}{2} + q(r+1).$$

Substituting $r = 0, 1, 2$ gives the three displayed cases. □

Remark 3.3. Combined with the elementary fact that $\lfloor n^2/2 \rfloor$ depends only on the parity of n, Theorem 3.2 allows $T(n)$, under Conjecture 3.1, to be written as an explicit quadratic polynomial in t once the residue of n modulo 6 is fixed. For instance, writing $n = 6t + r'$ for $r' = 0, \ldots, 5$ gives $T(n) = 24t^2 + c_1(r')t + c_0(r')$ for explicit constants; e.g. $T(6t) = 24t^2 - t$ and $T(6t+1) = 24t^2 + 7t$. We omit the routine remaining cases.

Corollary 3.4. *Under Conjecture 3.1, $T(n) = \frac{2}{3}n^2 + O(n)$.*

4. A proven lower bound and an exact accounting identity

Definition 4.1. For a configuration σ, let $D(\sigma) = \sum_{\text{balls}} |\,\text{current bin} - \text{target bin}\,|$, the total displacement.

Lemma 4.2. *$D(\text{start}) = 2\lfloor n^2/2 \rfloor$, and $D(\text{target}) = 0$. Every move changes D by exactly -2, 0, or $+2$.*

Proof. $D(\text{start}) = 2\sum_{i=1}^{n} |2i - n - 1|$, a standard sum equal to $2\lfloor n^2/2 \rfloor$. A move exchanges one ball from bin i with one from bin $i+1$; each of the two balls moves exactly one bin, so its individual contribution to D changes by ± 1 according to whether it moves toward or away from its target (or, if already at its target, necessarily away, i.e. $+1$). Summing the two independent ± 1 contributions gives a total change in $\{-2, 0, +2\}$. □

Call a move *useful* if D decreases by 2, *neutral* if D is unchanged, and *regressive* if D increases by 2.

Theorem 4.3 (Exact accounting identity)**.** *Let a complete sequence of moves from start to target use u useful, z neutral, and p regressive moves. Then*

$$u - p = \left\lfloor \frac{n^2}{2} \right\rfloor \qquad \text{and} \qquad T = u + z + p = \left\lfloor \frac{n^2}{2} \right\rfloor + z + 2p,$$

where T denotes the length of the sequence. In particular this holds for every *complete sequence, not only an optimal one.*

Proof. Telescoping D along the sequence gives $D(\text{target}) - D(\text{start}) = -2u + 2p$, and by Lemma 4.2 this equals $-2\lfloor n^2/2 \rfloor$. Solving gives $u - p = \lfloor n^2/2 \rfloor$, and substituting into $T = u + z + p$ gives the second identity. □

Corollary 4.4. *$T(n) \geq \lfloor n^2/2 \rfloor$ for all n.*

Theorem 4.3 reduces Conjecture 3.1 to a single strategy-independent inequality.

Proposition 4.5. *Conjecture 3.1 is equivalent to the assertion that every complete sequence of moves satisfies*

$$z + 2p \;\geq\; S(n).$$

We were unable to prove Proposition 4.5's inequality in general; the remainder of the paper documents both partial progress toward it and a substantial catalogue of approaches that fail.

We do, however, have one further elementary lower bound, sufficient to slightly strengthen Corollary 4.4 for odd n.

Lemma 4.6. *For odd $n \geq 3$, no move available from the start configuration is useful. Consequently $T(n) \geq \lfloor n^2/2 \rfloor + 1$ for odd $n \geq 3$.*

Proof. At the start configuration, bin i contains label $n + 1 - i$. A move at boundary i is useful precisely when bin i contains a ball wanting to move right (target bin $> i$, i.e. $n + 1 - i > i$, i.e. $i < \frac{n+1}{2}$) and bin $i + 1$ contains a ball wanting to move left (target bin $< i + 1$, i.e. $n - i < i + 1$, i.e. $i > \frac{n-1}{2}$). Both conditions hold simultaneously only for i in the open interval $\left(\frac{n-1}{2}, \frac{n+1}{2}\right)$, an interval of length 1. When n is odd, both endpoints are integers, so the open interval contains no integer, and no boundary is useful. Hence the first move of any sequence is neutral or regressive, contributing at least 1 to $z + 2p$, and the claim follows from Theorem 4.3. □

Remark 4.7. The corresponding computation for even n shows exactly one boundary, $i = n/2$, is useful at the start. This asymmetry between odd and even n does not, by itself, explain the full magnitude of $S(n)$, which grows quadratically; see the discussion in Section 6.

5. A reduced-word reformulation, and its limits

In this section we develop an exact reformulation of the problem in the language of reduced words for the longest element of a symmetric group, prove a genuinely new lemma within that framework, and then explain, with a supporting negative result, why this framework — at least via the routes we have explored — does not by itself resolve Proposition 4.5.

5.1. **The correspondence.** Let $N = 2n$. Assign to the two balls initially in bin i the *wire labels* $2i - 1$ and $2i$ (in either order). A configuration is then equivalent to an assignment of the N wires to N *slots* $1, \ldots, N$, where slots $2i - 1, 2i$ constitute bin i, together with a choice, for each bin, of which of its two wires occupies the lower-numbered slot; the latter choice carries no physical meaning; it is a bookkeeping convention.

Proposition 5.1. *A real move at boundary i corresponds to using the transposition (Coxeter generator) s_{2i} of S_N exactly once, optionally preceded and/or followed by transpositions s_{2i-1} or s_{2i+1} (relabelings within a bin, which are not physical moves and cost nothing). Consequently, writing $H = \langle s_1, s_3, \ldots, s_{N-1} \rangle \cong (\mathbb{Z}/2)^n$ for the abelian group generated by the odd-indexed generators, we have*

$$T(n) = c(w_0),$$

where $w_0 \in S_N$ is the longest element (full reversal, $w_0(p) = N + 1 - p$), and $c(w_0)$ denotes the minimum number of even-indexed generators occurring in any expression for w_0 as a word in $s_1, \ldots, s_{N-1}$ that is reduced when the odd-indexed letters are freely discarded.

Proof sketch. The claim that a real move corresponds to exactly one even generator, freely surrounded by odd generators, follows from the canonicalization described above. The set of "bin-respecting" target permutations (those compatible with the target configuration under some choice of within-bin orderings) forms a single double coset Hw_0H, since all such permutations induce the same map on bins (full reversal) and H acts by independently permuting within each bin. Because H-factors can be freely inserted immediately before and after any real move (this uses that H is generated by pairwise-commuting involutions with disjoint supports), the minimum cost $c(\rho)$ is constant across the double coset Hw_0H, and equals $c(w_0)$. □

We independently verified Proposition 5.1 by computing $c(w_0)$ via a dynamic program over the weak order of S_N — a computation entirely disjoint from the bin-configuration search of Section 2 — and found exact agreement with Table 1 for $n = 1, \ldots, 5$.

We use the following standard fact without proof (it follows from the exchange condition for Coxeter groups, or directly by induction on word length).

Proposition 5.2 (Standard)**.** *In any reduced word for $w_0 \in S_N$, every pair of indices $a < b$ has its wires cross exactly once.*

5.2. **A new lemma on three-wire crossings.**

Lemma 5.3 (Triple sandwiching)**.** *Let $a < b < c$ be wire indices. In any reduced word for w_0, the crossing of wires a and c occurs, in time, strictly between the crossings of a, b and of b, c.*

Proof. The relative order of $\{a, b, c\}$ among themselves changes only at crossings involving two of these three wires; crossings involving an outside wire leave their mutual order unaffected. Restricted to these three wires, the induced sub-word is therefore a reduced word for the reversal of (a, b, c), i.e. for the longest element of S_3, whose only two reduced words are $s_1s_2s_1$ and $s_2s_1s_2$. Translating back, these correspond to the crossing orders $(a, b), (a, c), (b, c)$ and $(b, c), (a, c), (a, b)$ respectively; in both, (a, c) occurs strictly between the other two. □

We verified Lemma 5.3 computationally against every triple of wires in the optimal reduced word for w_0 (for $n = 2, 3, 4$) and additionally against a deliberately *non*-optimal reduced word (the standard bubble-sort word, for $n = 2, \ldots, 5$), finding zero violations in every case, confirming that the lemma is a property of *all* reduced words, not merely optimal ones.

5.3. **Why this framework alone does not suffice.** The cost function $c(w_0)$ depends on the *position* (which generator) at which each crossing occurs, whereas Lemma 5.3 constrains only the *temporal order* of crossings. We investigated, and rule out, two natural ways of converting temporal information into positional information.

Proposition 5.4. *It is not true that, in a reduced word for w_0, the position $\kappa(a, c)$ of the crossing of a, c lies between $\kappa(a, b)$ and $\kappa(b, c)$ for all $a < b < c$; nor is it true that $\kappa(a, c) = \kappa(a, b) + \kappa(b, c)$.*

Evidence. For $N = 4$, the optimal reduced word yields $\kappa(1, 2) = 1, \kappa(2, 3) = 1, \kappa(1, 3) = 2$ (already outside the interval $[1, 1]$, refuting betweenness on all four triples of the smallest nontrivial case) while coincidentally satisfying additivity. Additivity itself fails for $N = 6$: only 5 of the 20 triples satisfy $\kappa(a, c) = \kappa(a, b) + \kappa(b, c)$, and the fraction does not improve at $N = 8, 10$. □

Proposition 5.5. *For $A_b := \#\{a < b < c : \kappa(a, c)$ is even$\}$, there is no universal constant $L_b > 0$ with $A_b \geq L_b$ for every reduced word.*

Evidence. For $N = 4$, the reduced word $[(2, 2, 3), (1, 1, 3), (2, 1, 2), (3, 1, 4), (2, 2, 4), (1, 3, 4)]$ (generator index, wire, wire) achieves $A_2 = 0$. □

We regard Propositions 5.4 and 5.5 as establishing that Lemma 5.3, while a genuine and previously unrecorded structural fact about reduced words for w_0, does not by itself carry enough

positional information to prove Proposition 4.5. We were also unable to obtain any nontrivial bound from single-wire arguments alone.

Proposition 5.6. *Using only the constraint that wire k (for $1 \le k \le N$) participates in exactly $N-k$ rightward and $k-1$ leftward crossings (forced by its endpoints $k \to N+1-k$), the best lower bound obtainable on $c(w_0)$ is exactly $\lfloor n^2/2 \rfloor$, recovering Corollary 4.4 and nothing more.*

6. Further ruled-out approaches

We record here, in the interest of sparing future investigators duplicated effort, several further natural strategies that we investigated and rejected, in each case via an explicit witness.

Proposition 6.1 (Local potentials do not suffice)**.** *Let Φ be any linear combination of the following 34 features of a configuration: counts of each of the six possible bin "type" patterns (home/home, right/right, left/left, right/left, right/home, left/home) refined by bin-index residue mod 3; counts of blocked boundaries refined by boundary-index residue mod 3; a histogram of ball-displacement residues mod 3; and counts of rightward/leftward-active bins refined by parity of bin index. Requiring $\Phi(ns)-\Phi(s) \le w(move)$ (with $w = 0, 1, 2$ for useful, neutral, regressive moves respectively) for* every *directed edge of the full reachable configuration graph, the maximum provable value of $\Phi(target) - \Phi(start)$ is exactly 0, for $n = 3, 4, 5$, including after removing all regressive edges from the constraint set.*

Method. Formulated as a linear program (features as columns, one constraint per directed edge of the exhaustively enumerated reachable graph) and solved with a standard LP solver; verified the result is not an artifact of coefficient bounds by re-solving with bounds enlarged by four orders of magnitude, obtaining the same value. □

Proposition 6.2 (Pure local greedy is not optimal)**.** *The rule "if any useful move is available, perform one (arbitrary tie-break), otherwise perform an arbitrary move" does not reliably reach $T(n)$ for $n \ge 5$: in 20 independent random tie-breaking trials each for $n = 5, 6$, the rule never once matched the true optimum (best of 20 trials: 20 vs. true value 15 for $n = 5$; 29 vs. true value 23 for $n = 6$).*

Proposition 6.3 (Bin-removal induction fails)**.** *It is not true that an optimal solution for n bins restricts, on the "inner" bins $2, \dots, n-1$, to an optimal (or even a valid, self-contained) solution for the corresponding $(n-2)$-bin problem. In an optimal solution for $n = 6$, a ball whose label belongs to an inner bin is observed to visit an outer bin (bin 1) at an intermediate step.*

Proposition 6.4 (Waste is not equivalent to total blockage)**.** *It is not true that every neutral or regressive move occurs only when* no *boundary admits a useful move. In the optimal solutions found for $n = 4$ and $n = 6$, several neutral moves occur while a useful move is available at another boundary.*

We also isolated a precise mechanism explaining, in part, why Proposition 6.1 must fail for *any* choice of bin-type-count features, however refined.

Lemma 6.5 (Type preservation)**.** *Suppose bins $i, i+1$ both consist of two balls each wanting to move right (a "pure-RR" pair). A move transferring a ball from bin $i+1$ to bin i always leaves bin i pure-RR. A move transferring a ball from bin i to bin $i+1$ leaves bin $i+1$ pure-RR unless the transferred ball's target is exactly bin $i+1$, in which case bin $i+1$ becomes mixed. The symmetric statement holds for adjacent pure-LL bins.*

Proof. If bin $i+1$ is pure-RR, every ball in it has target $> i+1$, hence target $> i$ as well; such a ball, moved to bin i, still wants to move right. This proves the first claim. For the second, a ball in bin i with target $> i$ moved to bin $i+1$ either still has target $> i+1$ (remains right-wanting) or has target exactly $i+1$ (becomes home). □

Lemma 6.5 shows that certain neutral moves change *which specific labels* occupy a bin while leaving its type profile (home/right/left) entirely unchanged; such moves are, by construction, invisible to any potential built from bin-type counts, however finely refined by residues, which is precisely the obstruction witnessed numerically in Proposition 6.1.

7. An auxiliary wall-crossing problem

Motivated by Lemma 6.5, we isolated a smaller, precisely defined subproblem intended to capture the cost of routing traffic across a block of otherwise-resolved bins.

Definition 7.1. For $L \geq 0$, the *wall problem of length* L has $L+2$ bins. Bin 0 starts with two balls labeled A (target: bin $L+1$); bin $L+1$ starts with two balls labeled B (target: bin 0); each bin j, $1 \leq j \leq L$, starts with two balls labeled j, with target bin j (i.e. already correctly placed). Let $T_{\text{wall}}(L)$ be the minimum number of moves to reach the configuration with bin $0 = \{B, B\}$, bin $L+1 = \{A, A\}$, and every middle bin restored to $\{j, j\}$.

Theorem 7.2. *$T_{\text{wall}}(0) = 2$, $T_{\text{wall}}(1) = 5$, and $T_{\text{wall}}(L) = 4L$ for all $L \geq 2$.*

We verified Theorem 7.2 by exact (bidirectional) breadth-first search for $L = 0, \ldots, 6$, using two independent implementations (Python and C++, the latter using open-addressing hash tables for memory efficiency), which agreed exactly. We regard the formula as established with very high confidence for $L \leq 6$; we have not proved it for general L, though the pattern is clean and the case $L \geq 2$ is linear with no further irregularity across five consecutive confirmed values.

Observation 7.3. The optimal move sequence found for $L = 3$ has a striking two-phase structure: the first 4 moves (exactly $2L-2$) are all neutral, achieved via successive full pairwise exchanges of adjacent bins' contents; the remaining 8 moves (exactly the displacement baseline $2(L+1)$) are all useful. This priming-then-useful pattern was also visible, qualitatively, in the smallest optimal solutions of the main problem itself ($n = 3, 4, 5$).

7.1. A nested nested experiment and an unresolved discrepancy at $n = 8$.

We generalized Definition 7.1 by replacing the L trivially-home middle bins with an active size-m reversal subinstance (for $1 \leq m \leq L$), occupying the first m of the L gap bins, with any remaining $L - m$ bins trivial as before. Writing $\text{excess}(L, m)$ for the resulting excess above the displacement baseline, we computed $\text{excess}(L, m)$ exactly for all $0 \leq m \leq L \leq 5$. Two consistency checks passed in every case: $\text{excess}(L, 0) = \text{excess}(L, 1) = g(L) := T_{\text{wall}}(L) - 2(L+1)$ (a size-1 reversal is trivial), and $\text{excess}(L, L) = S(L+2)$, recovering the known values of Table 1 exactly, as it must since $m = L$ reduces the construction to the original $(L+2)$-bin problem.

Define $\text{additional}(L) := S(L+2) - S(L)$, the marginal excess attributable to the outer pair crossing an *active* interior, using only already-verified data (no extrapolation).

Observation 7.4 (50% absorption)**.** For $L = 3, 4, 5$, $\text{additional}(L) = L - 1$, exactly half of $g(L) = 2L - 2$. (For $L = 1, 2$, $\text{additional}(L) = g(L)$, i.e. no absorption, consistent with the interior having negligible structure at these sizes.)

Observation 7.4 is a clean, striking pattern, and we have a plausible qualitative explanation for it (roughly: when the interior has its own genuine task, a middle ball displaced in the direction it already needed to travel costs nothing extra, while displacement in the opposite direction remains pure waste, suggesting a natural 50% split) but no proof.

More importantly, Observation 7.4, if extrapolated one further step to $L = 6$ (i.e. $n = 8$), predicts

$$\text{additional}(6) = 5 \implies S(8) = S(6) + 5 = 5 + 5 = 10,$$

whereas Conjecture 3.1, via the closed form of Theorem 3.2, predicts

$$S(8) = S(7) + \left\lfloor \tfrac{8}{3} \right\rfloor = 7 + 2 = 9.$$

Open Problem 7.5. Determine $T(8)$ (equivalently $S(8)$) independently, and thereby resolve whether $S(8) = 9$ (consistent with Conjecture 3.1) or $S(8) = 10$ (consistent with a naive extrapolation of Observation 7.4), or neither.

We regard Open Problem 7.5 as the single most valuable immediate next computation for this line of work: both live hypotheses are supported by exact, non-extrapolated data at every point checked so far, and they make different, mutually exclusive predictions at the very next value. We emphasize that Observation 7.4 is exactly that — an observation drawn from five confirmed data points ($L = 1, \ldots, 5$) — and not a theorem; we present the resulting tension as a demonstration of the danger of extrapolating even a very clean-looking finite pattern, rather than as evidence against Conjecture 3.1, which remains, in our judgment, the better-supported of the two hypotheses by virtue of resting on seven confirmed values rather than five.

8. Discussion and open problems

We summarize the logical status of every claim in this paper, to guard against the conjectural and the proven being conflated.

- **Proved**: Table 1 (exhaustive search); Theorem 3.2 (closed form for $S(n)$); Lemma 4.2, Corollary 4.4, Theorem 4.3, Proposition 4.5 (the baseline bound and accounting identity); Lemma 4.6 (odd-n strengthening); Proposition 5.1 and Lemma 5.3 (the reduced-word reformulation and the triple-sandwiching lemma); Lemma 6.5 (type preservation); Propositions 5.4, 5.5, 6.1, 6.2, 6.3, 6.4 (each a rigorously witnessed negative/insufficiency result).
- **Exhaustively verified for stated small cases, not proved in general**: Conjecture 3.1 ($n \leq 7$); Theorem 7.2 ($L \leq 6$).
- **Observed pattern from exact data, explicitly not claimed as a theorem**: Observation 7.4, whose naive extrapolation conflicts with Conjecture 3.1 at $n = 8$ (Open Problem 7.5).

The central open problem remains proving (or disproving) Proposition 4.5's inequality $z + 2p \geq S(n)$ for every complete sequence of moves. We list what we regard as the most promising remaining directions, in rough order of how much of this paper's machinery they would reuse.

Open Problem 8.1. Find a strategy-independent argument — not resting on any single-wire, aggregate-parity, or local-potential technique, all of which are ruled out by Section 6 and Section 5 — proving $z + 2p \geq S(n)$.

Open Problem 8.2. Prove Theorem 7.2 for general L (we have only verified it for $L \leq 6$), and determine whether the nested generalization of Section 7 admits an exact recursive formula for $\text{excess}(L, m)$.

Open Problem 8.3. Resolve Open Problem 7.5. More generally, determine whether $S(n) - S(n-2)$ follows the sum-of-floors pattern of Conjecture 3.1 or the linear pattern suggested by Observation 7.4, or discover the correct formula reconciling both bodies of evidence.

Open Problem 8.4. Determine whether $p = 0$ in every optimal solution (observed in every optimal trace examined in this work, for $n \leq 6$, but not proved in general).

Acknowledgments

We thank Peter Winkler for posing this puzzle in [1], and acknowledge the methodological debt this work owes to [2] for demonstrating that a puzzle from Winkler's "Unsolved Puzzles" chapter can be fully resolved by a combination of careful combinatorial structure and computation; we regret that we have, so far, only partially succeeded in following that example.

References

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[2] T. V. Madhavarapu, *The Most Malicious Maître D'*, arXiv:2407.09000 (2025).